Fuse Machines
AI Fellowship – 2026

Assignment 1 – Part 2
SQL

Submitted By:

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Task:

1. Show all the customers whose creditLimit is greater than 20000

Showing rows 0 - 24 (97 total, Query took 0.0003 seconds.)

```
select customerName,creditLimit from customers where creditLimit>20000;
```

Profiling [Edit inline

1 > >> | Show all | Number of rows: 25 Filter rows: Search this table

Extra options

←T→	customerName	creditLimit
Edit Copy Delete	Atelier graphique	21000.00
Edit Copy Delete	Signal Gift Stores	71800.00
Edit Copy Delete	Australian Collectors, Co.	117300.00
Edit Copy Delete	La Rochelle Gifts	118200.00
Edit Copy Delete	Baane Mini Imports	81700.00
Edit Copy Delete	Mini Gifts Distributors Ltd.	210500.00
Edit Copy Delete	Blauer See Auto, Co.	59700.00
Edit Copy Delete	Mini Wheels Co.	64600.00
Edit Copy Delete	Land of Toys Inc.	114900.00
Edit Copy Delete	Euro+ Shopping Channel	227600.00
Edit Copy Delete	Volvo Model Replicas, Co	53100.00
Edit Copy Delete	Danish Wholesale Imports	83400.00
Edit Copy Delete	Saveley & Henriot, Co.	123900.00

2. Show the employees who report to VP Sales.

Showing rows 0 - 3 (4 total, Query took 0.0004 seconds.)

```
select * from employees where reportsTo=1056;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTit
Edit Copy Delete	1088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1056	Sales Manag (APAC
Edit Copy Delete	1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1056	Sale Manag (EME/
Edit Copy Delete	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manag (NA)
Edit Copy Delete	1621	Nishi	Mami	x101	mnishi@classicmodelcars.com	5	1056	Sales Rep

Check all *With selected:* Edit Copy Delete Export

3. Find all the customers who have set their state while filling the forms and Lives in USA and credit limit is between 100000 and 200000.

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking M

Showing rows 0 - 7 (8 total, Query took 0.0004 seconds.)

```
select customerNumber,customerName,contactLastName,contactFirstName,city,state,country,creditLimit from customers where state is not
null and country='USA' and creditLimit between 100000 and 200000;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	customerNumber	customerName	contactLastName	contactFirstName	city	state	country	creditLimit
Edit Copy Delete	131	Land of Toys Inc.	Lee	Kwai	NYC	NY	USA	114900.00
Edit Copy Delete	151	Muscle Machine Inc	Young	Jeff	NYC	NY	USA	138500.00
Edit Copy Delete	157	Diecast Classics Inc.	Leong	Kelvin	Allentown	PA	USA	100600.00
Edit Copy Delete	239	Collectable Mini Designs Co.	Thompson	Valarie	San Diego	CA	USA	105000.00
Edit Copy Delete	286	Marta's Replicas Co.	Hernandez	Marta	Cambridge	MA	USA	123700.00
Edit Copy Delete	319	Mini Classics	Frick	Steve	White Plains	NY	USA	102700.00
Edit Copy Delete	321	Corporate Gift Ideas Co.	Brown	Julie	San Francisco	CA	USA	105000.00
Edit Copy Delete	363	Online Diecast Creations Co.	Young	Dorothy	Nashua	NH	USA	114200.00

4. Find all the employees who report to Sales Managers of all types.

Showing rows 0 - 14 (15 total, Query took 0.0004 seconds.)

```
select * from employees where reportsTo in (1088,1102,1143);
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
Edit Copy Delete	1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep
Edit Copy Delete	1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep
Edit Copy Delete	1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep
Edit Copy Delete	1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep
Edit Copy Delete	1286	Tseng	Foon Yue	x2248	ftseng@classicmodelcars.com	3	1143	Sales Rep
Edit Copy Delete	1323	Vanauf	George	x4102	gvanauf@classicmodelcars.com	3	1143	Sales Rep
Edit Copy Delete	1337	Bondur	Loui	x6493	lbondur@classicmodelcars.com	4	1102	Sales Rep
Edit Copy Delete	1370	Hernandez	Gerard	x2028	ghernande@classicmodelcars.com	4	1102	Sales Rep
Edit Copy Delete	1401	Castillo	Pamela	x2759	pcastillo@classicmodelcars.com	4	1102	Sales Rep

5. Find the average credit limit of customers of each country.

Showing rows 0 - 24 (27 total, Query took 0.0004 seconds.)

```
select country,avg(creditLimit) as avg_creditLimit from customers group by country order by avg_creditLimit desc;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Crea

1 > >> | Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	country	avg_creditLimit 1
Edit Copy Delete	Denmark	102100.000000
Edit Copy Delete	Italy	97200.000000
Edit Copy Delete	Finland	95266.666667
Edit Copy Delete	Norway	91200.000000
Edit Copy Delete	New Zealand	90625.000000
Edit Copy Delete	UK	88740.000000
Edit Copy Delete	Japan	87800.000000
Edit Copy Delete	Australia	86060.000000
Edit Copy Delete	Sweden	84750.000000
Edit Copy Delete	Philippines	81500.000000
Edit Copy Delete	USA	78102.777778
Edit Copy Delete	France	77691.666667
Edit Copy Delete	Canada	76200.000000
Edit Copy Delete	Spain	73971.428571
Edit Copy Delete	Singapore	67233.333333

6. Find the total no. of orders for each date and customer. Show only dates with total number of orders greater than 10 for date and customer.

Showing rows 0 - 24 (135 total, Query took 0.0026 seconds.)

```
select o.orderDate,o.customerNumber,count(od.orderNumber) as total_orders, sum(od.quantityOrdered) as quantityOrdered from orders o
join orderdetails od on o.orderNumber=od.orderNumber group by orderDate,customerNumber having count(o.orderNumber)>10;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refre

1 > >> | Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

orderDate	customerNumber	total_orders	quantityOrdered
2003-01-29	121	16	541
2003-01-31	141	13	443
2003-02-11	145	15	545
2003-02-17	278	18	675
2003-03-03	385	16	561
2003-03-18	187	16	570
2003-04-16	148	12	402
2003-04-28	382	14	442
2003-04-29	114	15	525
2003-05-08	350	17	545
2003-05-21	112	13	448
2003-05-28	458	17	617
2003-06-03	151	15	540
2003-07-02	124	17	607
2003-07-07	496	13	420
2003-07-24	161	11	385

7. Find the name of the supervisor, job title of supervisor and total no. of supervisee using subquery. (Without using Join operation)

Showing rows 0 - 5 (6 total, Query took 0.0004 seconds.)

```
select e.lastName,e.firstName,e.jobTitle, (select count(*) from employees emp where emp.reportsTo = e.employeeNumber) as
total_supervisee from employees e having total_supervisee > 0 ORDER BY `total_supervisee` DESC;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

lastName	firstName	jobTitle	total_supervisee 1
Bondur	Gerard	Sale Manager (EMEA)	6
Bow	Anthony	Sales Manager (NA)	6
Patterson	Mary	VP Sales	4
Patterson	William	Sales Manager (APAC)	3
Murphy	Diane	President	2
Nishi	Mami	Sales Rep	1

8. Find the name of the supervisor, job title of supervisor and total no. of supervisee using subquery. (With using Join operation)

Showing rows 0 - 5 (6 total, Query took 0.0004 seconds.)

```
select e.lastName, e.firstName, e.jobTitle, count(emp.employeeNumber) as total_supervisee from employees e join employees emp on
e.employeeNumber = emp.reportsTo group by e.employeeNumber order by total_supervisee desc;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [R

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

lastName	firstName	jobTitle	total_supervisee 1
Bondur	Gerard	Sale Manager (EMEA)	6
Bow	Anthony	Sales Manager (NA)	6
Patterson	Mary	VP Sales	4
Patterson	William	Sales Manager (APAC)	3
Murphy	Diane	President	2
Nishi	Mami	Sales Rep	1

9. Find all customers with a credit limit greater than average credit limit using WITH Clause.

Showing rows 0 - 24 (69 total, Query took 0.0004 seconds.)

```
with AvgCredit as (select avg(creditLimit) as avg_credit from customers) select c.customerName, c.creditLimit from customers c,
AvgCredit ac where c.creditLimit > ac.avg_credit;
```

[Edit inline] [Edit] [Create P

1 > >> | Show all | Number of rows: 25 Filter rows: Search this table

Extra options

	customerName	creditLimit
Edit Copy Delete	Signal Gift Stores	71800.00
Edit Copy Delete	Australian Collectors, Co.	117300.00
Edit Copy Delete	La Rochelle Gifts	118200.00
Edit Copy Delete	Baane Mini Imports	81700.00
Edit Copy Delete	Mini Gifts Distributors Ltd.	210500.00
Edit Copy Delete	Land of Toys Inc.	114900.00
Edit Copy Delete	Euro+ Shopping Channel	227600.00
Edit Copy Delete	Danish Wholesale Imports	83400.00
Edit Copy Delete	Saveley & Henriot, Co.	123900.00
Edit Copy Delete	Dragon Souveniers, Ltd.	103800.00
Edit Copy Delete	Muscle Machine Inc	138500.00
Edit Copy Delete	Diecast Classics Inc.	100600.00
Edit Copy Delete	Technics Stores Inc.	84600.00
Edit Copy Delete	Handji Gifts& Co	97900.00
Edit Copy Delete	Herkku Gifts	96800.00

10. Find the rank of customer. [Customer with highest credit limit have 1 rank and Customer with lowest credit limit have highest rank]. Then, find the customer with the third highest credit limit.

Showing rows 0 - 0 (1 total, Query took 0.0016 seconds.)

```
WITH RankedCustomers AS ( SELECT customerName, creditLimit, RANK() OVER (ORDER BY creditLimit DESC) AS customer_rank FROM customers )
SELECT * FROM RankedCustomers WHERE customer_rank = 3;
```

[Edit inline] [Edit] [Create PHP co

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

	customerName	creditLimit	customer_rank
Edit Copy Delete	Vida Sport, Ltd	141300.00	3

Check all *With selected:* Edit Copy Delete

11. Generate a report that shows total no. of employees working in each office.

Showing rows 0 - 6 (7 total, Query took 0.0009 seconds.)

```
select o.city,o.officeCode,count(e.employeeNumber) as total_emp from offices o join employees e on
o.officeCode=e.officeCode group by o.officeCode, o.city order by total_emp desc;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25

Extra options

city	officeCode	total_emp 1
San Francisco	1	6
Paris	4	5
Sydney	6	4
Boston	2	2
NYC	3	2
Tokyo	5	2
London	7	2

12. Generate a report that shows total no. of customers visited each office.

Showing rows 0 - 6 (7 total, Query took 0.0006 seconds.)

```
select o.city,o.officeCode,count(c.customerNumber) as total_customers from offices o join
employees e on o.officeCode=e.officeCode join customers c on e.employeeNumber =
c.salesRepEmployeeNumber group by o.officeCode, o.city order by total_customers desc;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Re

Show all | Number of rows: 25

Extra options

city	officeCode	total_customers 1
Paris	4	29
London	7	17
NYC	3	15
San Francisco	1	12
Boston	2	12
Sydney	6	10
Tokyo	5	5

13. Generate a report that shows total payment received by each office using payment tables and essential tables. The report should show the office name, state and country, along with total payments made.

Showing rows 0 - 6 (7 total, Query took 0.0008 seconds.)

```
select o.city as office_name, o.state, o.country, sum(p.amount) as total_payment from offices o
join employees e on o.officeCode = e.officeCode join customers c on e.employeeNumber =
c.salesRepEmployeeNumber join payments p on c.customerNumber = p.customerNumber group by o.city
order by total_payment desc;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refres

Show all | Number of rows: 25

Extra options

office_name	state	country	total_payment 1
Paris	*NULL*	France	2819168.90
San Francisco	CA	USA	1337439.58
London	*NULL*	UK	1324325.90
NYC	NY	USA	1072619.47
Sydney	*NULL*	Australia	1007292.98
Boston	MA	USA	835882.33
Tokyo	Chiyoda-Ku	Japan	457110.07

14. Generate a report that shows total sales(in amount) by each office using order details table and other essential tables.

Showing rows 0 - 6 (7 total, Query took 0.0042 seconds.)

```
select o.city, o.officeCode, o.country, sum(od.quantityOrdered * od.priceEach) as total_sales from
offices o join employees e on o.officeCode = e.officeCode join customers c on e.employeeNumber =
c.salesRepEmployeeNumber join orders ord on c.customerNumber = ord.customerNumber join
orderdetails od on ord.orderNumber = od.orderNumber group by o.officeCode,o.city order by
total_sales desc;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh

Show all | Number of rows: 25

Extra options

city	officeCode	country	total_sales 1
Paris	4	France	3083761.58
London	7	UK	1436950.70
San Francisco	1	USA	1429063.57
NYC	3	USA	1157589.72
Sydney	6	Australia	1147176.35
Boston	2	USA	892538.62
Tokyo	5	Japan	457110.07

15. Generate a report that shows total payment pending for each office.

Showing rows 0 - 6 (7 total, Query took 0.0051 seconds.)

```
with officeSales as (select o.city, o.officeCode, o.country, sum(od.quantityOrdered * od.priceEach) as total_sales from offices o join employees e on o.officeCode = e.officeCode join customers c on
e.employeeNumber = c.salesRepEmployeeNumber join orders ord on c.customerNumber = ord.customerNumber join orderdetails od on ord.orderNumber = od.orderNumber group by o.officeCode,o.city), officePayments
as (select o.officeCode, sum(p.amount) as total_payment from offices o join employees e on o.officeCode = e.officeCode join customers c on e.employeeNumber = c.salesRepEmployeeNumber join payments p on
c.customerNumber = p.customerNumber group by o.officeCode) select os.city as office_name, os.total_sales,op.total_payment, (os.total_sales - op.total_payment) as total_pending from officeSales os join
officePayments op on os.officeCode = op.officeCode order by total_pending desc;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

office_name	total_sales	total_payment	total_pending
Paris	3083761.58	2819168.90	264592.68
Sydney	1147176.35	1007292.98	139883.37
London	1436950.70	1324325.90	112624.80
San Francisco	1429063.57	1337439.58	91623.99
NYC	1157589.72	1072619.47	84970.25
Boston	892538.62	835882.33	56656.29
Tokyo	457110.07	457110.07	0.00

16. Find the creditLimit of each person, proportion of creditLimit of each person in each country. [Proportion of person in USA = creditLimit of that person / sum (creditLimit) of all person in USA]

Showing rows 0 - 24 (98 total, Query took 0.0007 seconds.)

```
select customerName, country, creditLimit, sum(creditLimit) over(partition by country) as total_country_credit, (creditLimit/sum(creditLimit) over(partition by country)) as proportion from customers where creditLimit > 0 order by proportion desc;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh

1 > >> | Show all | Number of rows: 25

Extra options

←T→	customerName	country	creditLimit	total_country_credit	proportion 1
Edit Copy Delete	Clover Collections, Co.	Ireland	69400.00	69400.00	1.000000
Edit Copy Delete	Cruz & Sons Co.	Philippines	81500.00	81500.00	1.000000
Edit Copy Delete	Vida Sport, Ltd	Switzerland	141300.00	141300.00	1.000000
Edit Copy Delete	King Kong Collectables, Co.	Hong Kong	58600.00	58600.00	1.000000
Edit Copy Delete	Petit Auto	Belgium	79900.00	103400.00	0.772727
Edit Copy Delete	Scandinavian Gift Ideas	Sweden	116400.00	169500.00	0.686726
Edit Copy Delete	Salzburg Collectables	Austria	71700.00	117000.00	0.612821
Edit Copy Delete	Heintze Collectables	Denmark	120800.00	204200.00	0.591577
Edit Copy Delete	Tokyo Collectables, Ltd	Japan	94400.00	175600.00	0.537585

17. Create a view showing the customer name, complete address, and their total number of orders.

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0029 seconds.)

```
create view customer_summary as select c.customerName, concat_ws(', ', c.addressLine1, c.addressLine2, c.city, c.state, c.postalCode, c.country) as complete_address, count(o.orderNumber) as total_orders from customers c left join orders o on c.customerNumber = o.customerNumber group by c.customerNumber;
```

[Edit inline] [Edit] [Create PHP code

Showing rows 0 - ... (Query took 0.0008 seconds.)

```
select * from customer_summary;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP

> | Show all | Number of rows: 25

Extra options

customerName	complete_address	total_orders
Atelier graphique	54, rue Royale, Nantes, 44000, France	3
Signal Gift Stores	8489 Strong St., Las Vegas, NV, 83030, USA	3
Australian Collectors, Co.	636 St Kilda Road, Level 3, Melbourne, Victoria, 3...	5
La Rochelle Gifts	67, rue des Cinquante Otages, Nantes, 44000, Franc...	4
Baane Mini Imports	Erling Skakkes gate 78, Stavern, 4110, Norway	4
Mini Gifts Distributors Ltd.	5677 Strong St., San Rafael, CA, 97562, USA	17
Havel & Zbyszek Co	ul. Filtrowa 68, Warszawa, 01-012, Poland	0
Blauer See Auto, Co.	Lyonerstr. 34, Frankfurt, 60528, Germany	4
Mini Wheels Co.	5557 North Pendale Street, San Francisco, CA, 9421...	3
Land of Toys Inc.	897 Long Airport Avenue, NYC, NY, 10022, USA	4
Euro+ Shopping Channel	C/ Moralzarzal, 86, Madrid, 28034, Spain	26
Volvo Model Replicas, Co	Berguvsvägen 8, Luleå, S-958 22, Sweden	4
Danish Wholesale Imports	Vinbæltet 34, Kobenhavn, 1734, Denmark	5
Saveley & Henriot, Co.	2, rue du Commerce, Lyon, 69004, France	3
Dragon Souveniers, Ltd.	Bronz Sok., Bronz Apt. 3/6 Tesvikiye, Singapore, 0...	5
Muscle Machine Inc	4092 Furth Circle, Suite 400, NYC, NY, 10022, USA	4

18. Update the country of a customer (use any one record).

1 row affected. (Query took 0.0004 seconds.)

```
update customers set country = 'Spain' where customerNumber = 103;
```

[Edit inline] [E

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country
103	Atelier graphique	Schmitt	Carine	40.32.2555	54, rue Royale	NULL	Nantes	NULL	44000	Spain

19. Delete all payments below 20,000.

78 rows deleted. (Query took 0.0020 seconds.)

```
delete from payments where amount < 20000;
```

[Edit inline] [Edit]

20. Add new payments manually for an existing customer.

1 row inserted. (Query took 0.0003 seconds.)

```
insert into payments (customerNumber,checkNumber,paymentDate,amount) values (103,'HQ444443','2026-05-04',7000);
```

[Edit inline] [Edit] [Create PHP cod

customerNumber	checkNumber	paymentDate	amount
103	HQ444443	2026-05-04	7000.00

2 rows inserted. (Query took 0.0004 seconds.)

```
insert into payments (customerNumber,checkNumber,paymentDate,amount) values (103,'HQ444448','2026-05-04',7000),(103,'HQ442244','2026-05-05',10000);
```